Control Systems

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Control Systems (A09890)

Short Title: Control Systems
Faculty Science and Computing
Department Science
Cluster Mathematics and Physics
Author CWALSH
Credits 5 **Level:** Advanced

Description of Module / Aims

The module is designed for physics undergraduate students. There will be a comprehensive study of the concepts of control system theory as they have been developed in the frequency and time domains. The classical methods of control engineering are thoroughly covered in this module. The laboratory programme will be based on the use of the MATLAB and Simulink in the modelling, design and analysis of control systems.

Programmes

PHYI-0031	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	4 / 7 / M

Indicative Content

- System modelling: modelling physical dynamic systems, free body diagrams, analogous electrical and mechanical systems; differential equations, modelling in the frequency domain, Laplace transforms
- Time response: pole and zero plots, transient response of first-order systems; general response of second-order systems to step, ramp, impulse and sinusoidal input; under, critically and overdamped systems; performance of a step response of a second order under-damped system; approximation of higher-order systems
- Modelling in the time domain: state-space representation, electrical and mechanical systems in state space, transfer function to state space
- Reduction of multiple sub-systems: block representation, block diagram reduction of multiple subsystems; conversion from block diagrams to signal-flow diagrams; representation of state equations as signal-flow graphs
- Frequency response methods: frequency response plots and measurements; performance specifications in the frequency domain. Log magnitude and phase diagrams
- Control system design: stability criteria and analysis; PID controllers: design techniques, applications

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Construct the differential equations that describe the dynamic behaviour of physical systems, describe quantitatively the transient response of first-order and second order systems to unit step, ramp, impulse and sinusoidal inputs.
2. Derive transfer functions of dynamical systems using Laplace transformations.
3. Construct block diagram, and signal flow graph representations of systems using standard techniques.
4. Develop models of a range of systems in state space.
5. Determine the time response of a linear system and its dependence on the poles and zeros of the transfer function and approximate higher-order systems and systems using the dominant pole method.
6. Construct and interpret Bode frequency response plots and describe how the shape of these plots depend on the poles and zeros of the transfer function .
7. Design Simulink models of a range of dynamical systems.
8. Design MATLAB programs to generate root locus and other graphical aids and apply to PID tuning.

Learning and Teaching Methods

- Lectures: Lectures will be used to introduce new topics and their related concepts.
- Practicals: The practical element will be based on the use of MATLAB and Simulink in the modelling, design and analysis of control systems.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4,5,6
Continuous Assessment	30%	
Assignment	30%	7,8

Assessment Criteria

<40%: Student has demonstrated a very limited knowledge of control systems.

40%-49%: Student has demonstrated an adequate knowledge of control systems and is able to carry out some critical thinking of the subject material.

50%-59%: Shows a reasonable level of understanding of the various principles involved in control systems.

60%-69%: Shows good knowledge of the area. Shows ability to reason and analyze problems in the area of control system modelling analysis and design.

70%-100%: Excellent knowledge of the subject area. Excellent reasoning skills and focused well thought out arguments.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Supplementary Material

- Dorf, R. and R. Bishop. _Modern Control Systems_. 12th ed. United States of America: Prentice Hall, 2011.
- Golnaraghi, F. and B. Kuo. _Automatic Control Systems_. 9th ed. United States of America: Wiley, 2009.
- Klee, H. and R. Allen. _Simulation of Dynamic Systems with MATLAB and Simulink_. 2nd ed. United States of America: CRC Press, 2011.
- Nise, N. _Control Systems Engineering_. 6th ed. United States of America: Wiley, 2011.
- Pratap, R. _Getting Started with MATLAB: A Quick Introduction for Scientists and Engineers_. 1st ed. UK: Oxford University Press, 2009.
- Wilkie, J., M. Johnson and R. Katebi. _Control engineering an introductory course_. UK: Palgrave Macmillan, 2002.

Requested Resources

- SCIENCE LAB: Physics